

# Yuqun Wu

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Personal Website: <https://yuqunw.github.io>

## EDUCATION

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### University of Illinois at Urbana-Champaign

*Doctor of Philosophy in Computer Science*

- Advisor: Prof. Derek Hoiem

Champaign, USA

Aug 2023 - Present

*Master of Science in Computer Science (thesis)*

- Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang

Aug 2022 - May 2023

*Bachelor of Science in Computer Science & Statistics*

- Highest Honors at graduation, Dean's list for all years, GPA: 4.0/4.0

Jan 2020 - Dec 2021

### Sun Yat-sen University

*Bachelor of Science in Mathematics*

Guangzhou, China

Sep 2016 - Dec 2019

## RESEARCH EXPERIENCE

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### University of Illinois at Urbana-Champaign

Champaign, USA

#### SceneDiff: A Benchmark and Method for Multiview Object Change Detection

Jan 2025 - May 2025

Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - *ECCV 2026*

- Project targeted the problem of identifying regions that have changed between a pair of captures (images or videos) of the same scene at different times.
- Proposed a dataset and a new method for object change detection between a pair of captures (images or videos) of the same scene at different times.

#### TextRegion: Text-Aligned Region Tokens from Frozen Image-Text Models

Dec 2024 - May 2025

Advisor: Prof. Derek Hoiem - *TMLR 2025*

- Project proposed a training-free approach to enhance spatial understanding capacity of existing image-text models.
- Contributed to idea development, running experiments, and paper writing.

#### Region-based Representations Revisited

Sep 2023 - Nov 2023

Advisor: Prof. Derek Hoiem - *CVPR 2024*

- Project targeted at investigating new representation by combining SAM regions and dense features to solve various vision tasks, including semantic segmentation, object retrieval, video classification, and scene segmentation
- Responsible for implementation of feature extraction and pooling pipelines, and scene segmentation application on ScanNet

#### Improving Neural Radiance Fields with Patch-based Monocular Guidance

Jan 2023 - May 2023

Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - *3DV 2025*

- Project aimed to create 3D models that provide accurate geometry and view synthesis, partially closing the large geometric performance gap between NeRF and traditional MVS methods
- Proposed appearance regularization of normalized cross-correlation (NCC) and structural similarity (SSIM) between randomly sampled novel and training view to improve general performance

#### Plenoptic PNG: Real-Time Neural Radiance Fields in 150 KB

Aug 2022 - Dec 2023

Advisor: Prof. Derek Hoiem, Prof. Shenlong Wang - *3DV 2025*

- Project presented Quantized Fourier Features, which encoded a 3D scene into an extremely compact representation from 2D images and enabled its transmittance, decoding, and rendering in real-time across various platforms.
- Contributed to blending Quantized Fourier Features into different network setups, running experiments, and paper writing.

#### Sparse SPN: Depth Completion from Sparse Keypoints

Sep 2021 - Nov 2022

Advisor: Prof. Derek Hoiem

- Project targeted fast point clouds reconstruction from single view depth completion with SfM inputs.
- Proposed a novel method that outperforms existing depth completion pipelines given sparse keypoint depth, and reconstructed complete point clouds given SfM setup

## GRIT: General Robust Image Task Benchmark

*Jun 2021 - Aug 2021*

*Advisor: Prof. Derek Hoiem*

- Rendered surface normal of object-centric and scene-centric datasets, and split them into training, validation, and testing sets
- Trained a baseline network with training sets, and compare it with several other pretrained state-of-the-art normal estimation networks with testing sets
- Challenge Organizer of the 2nd workshop on Open World Vision of CVPR 2022

## University of California San Diego

Remote

### Lighting completion from sparse lighting samples

*Jun 2022 - Sep 2022*

*Advisor: Prof. Manmohan Chandraker*

- Project aims at recovering per-pixel spatially-varying lighting maps taking single color image and sparse lighting samples
- Investigated 2D lighting completion methods with differentiable rendering and compare to pure RGB-based estimation networks

## INDUSTRY EXPERIENCE

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### Research Scientist Internship

Redmond, USA

*Meta*

*May 2025 - Aug 2025*

## SERVICE

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### Conference Reviewer

Champaign, USA

*CVPR, ECCV, WACV*

*2024 - 2025*

### Teaching Assistant

Champaign, USA

*University of Illinois at Urbana-Champaign*

*Aug 2022 - May 2023*

- Course: *CS 445 Computational Photography, CS 441 Applied Machine Learning*

## SKILLS

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- **Programming Languages:** Python, C/C++, JavaScript, R
- **Other Tools:** Git, Pytorch, Latex